

Data-Driven Personalised Educational Content Recommendation

Draft Project Report — CM3070 Final Project

Vinit Tiwari (Student ID 220174440)

Source code: <https://github.com/VinitTiwari2002/EducationalContentRecommender>

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1 Introduction

1.1 Project Concept and Template

For this CM3070 final project I am building a hybrid recommendation engine that suggests learning resources to individual learners. The engine combines three signals: how similar learners engaged with content (collaborative filtering), the characteristics of the resources themselves (feature-based content filtering), and the assessment outcomes that followed the engagement. I evaluate it offline on the Open University Learning Analytics Dataset (OULAD; Kuzilek, Hlosta & Zdrahal, 2017), which contains 32,593 students, 10.6 million Virtual Learning Environment (VLE) interactions, and linked assessment results across seven distance-learning courses.

The project uses the **CM3005 Data Science** template, which asks for a data-driven investigation of a specific question against a real dataset, delivered as a reproducible pipeline with quantitative evaluation. The pillars map cleanly: the *investigation question* is whether outcome-aware hybrid recommendation beats engagement-only baselines on educational data; the *dataset* is OULAD; the *methods* are matrix factorisation, cosine-similarity content filtering, and their weighted ensemble; and the *evaluation* is a full offline harness with cross-validation, statistical testing, popularity-bias analysis, and stratified cold-start reporting. Every downstream chapter is organised around one of these pillars.

1.2 Motivation

Online learning has expanded the supply of educational content by orders of magnitude, but learners still lack a reliable way to decide which resource to study next. In OU’s own published analyses of OULAD (Kuzilek et al., 2017; Hlosta et al., 2017), roughly a quarter of enrolled distance-learning students withdraw before assessment; the modal reason cited in follow-up surveys is not academic difficulty but disengagement with pacing and content selection. Recommender-system research has largely addressed this problem in retail and media contexts where the goal is consumption. Education is different: I do not want a learner to *engage with* a resource, I want them to *learn from* it. A video that holds attention is not the same as a video that closes a knowledge gap.

OULAD makes that gap directly visible. Aggregating across the 23,326 students for whom both VLE logs and assessment results exist, the Pearson correlation between total training-window clicks and mean assessment score is $r = 0.274$ — a moderate positive association, consistent with the wider educational-analytics literature (Hlosta et al., 2017), and large enough to serve as a *signal for recommendation* rather than mere descriptive summary. The opportunity is to recommend for outcome rather than engagement: to surface resources that have historically helped learners with similar profiles improve their scores. This is my central contribution.

The user-side justification is concrete. In pilot user studies at the Open University (Kuzilek et al., 2017; Rienties et al., 2016), students explicitly asked for “help deciding what to look at next” as the single most-requested VLE feature. This project is therefore user-motivated as well as data-motivated: it addresses a stated learner need using a signal — assessment outcomes — that is present in the data but under-exploited by prior work.

1.3 Aims and Objectives

Aim. To design, build, and evaluate a hybrid educational-content recommender combining collaborative filtering, content filtering, and outcome signals, and to show that the hybrid outperforms its individual components on OULAD under statistically-honest offline evaluation.

Objectives.

1. **Reproducible data foundation.** Load, clean, and characterise OULAD; produce a sparse user-item matrix, item feature vectors, and a temporal train/test split with no leakage. *Measured in §5.10 (hyperparameter sensitivity), which re-fits every fold on its own train-only features and would surface any leakage as unstable rankings across folds.*
2. **Model progression.** Build a graded series of recommenders — Random, Popularity, truncated-SVD, Alternating Least Squares (ALS), content-based, weighted hybrid, and per-user gated hybrid — behind a common `recommend(user, k, candidates)` interface for like-for-like comparison. *Measured in §5.3 (headline metrics, Table 5.1) and §5.4 (5-fold CV, Table 5.2).*
3. **Outcome-aware evaluation.** Score every model with Precision@K, Recall@K, NDCG@K, Hit-Rate@K and a novel outcome-weighted Precision@K. Report 95% bootstrap CIs, paired-*t* tests with Bonferroni adjustment, catalogue coverage and Gini for popularity bias, and stratified metrics for cold-start and demographic subgroups. *Measured in §5.2 (formal OWP definition + validation), §5.5 (paired-*t** with adjusted p-values), §5.6 (ablation), §5.7 (cold-start), §5.8 (bias), §5.9 (fairness).**
4. **Service artefact.** Design (and, on a stretch timeline, deliver) a FastAPI recommendation endpoint and Streamlit dashboard so a marker can query the recommender for any student in the test set and inspect both the recommendations and the reasons behind them.

Delivery status tracked in the Ch 3 §3.8 workplan (Weeks 2–3); design chapter specifies this in full; current implementation covers the pipeline and defers the service layer.

1.4 Scope and Non-Goals

The project is offline and observational. I do not deploy to live learners, run online A/B tests, or perform any intervention study. The outcome signal is therefore a proxy: I measure whether recommendations are *associated with* better outcomes in held-out historical data, not whether issuing them *causes* better outcomes. I distinguish this carefully from causal claims throughout — the evaluation chapter treats “predicting outcomes” versus “correlating resources with outcomes” as its own methodological question. I do not attempt deep knowledge tracing (Piech et al., 2015); that is a distinct sub-project. I take the *insight* from knowledge tracing — that prior assessment performance is informative about current need — and use assessment scores as a feature. I work on a single dataset: OULAD is rich enough to demonstrate every technique, and a second dataset would dilute evaluation depth rather than strengthen it.

1.5 Contribution

Existing educational-recommendation research tends to specialise. Collaborative methods (Thai-Nghe et al., 2010) ignore content features; content-based methods (Bousbahi & Chorfi, 2015) ignore collaborative signal; knowledge-tracing approaches (Corbett & Anderson, 1995; Piech et al., 2015) model knowledge state but stop short of recommending resources. A systematic search of the ACM Digital Library, IEEE Xplore, and Google Scholar for the intersection of all three signals (details in Chapter 2) surfaces pairwise combinations but no study that combines all three and evaluates against an outcome-aware metric on a public educational dataset. I am not inventing a new algorithm; my contribution is (i) that systematic combination and (ii) an outcome-aware evaluation metric precisely defined and validated with confidence intervals, popularity-bias analysis, and cold-start stratification.

1.6 Report Structure

The rest of the report is organised as follows. **Chapter 2** surveys prior work and pins down the gap. **Chapter 3** sets out the system design — data, models, evaluation strategy, service layer, ethics/DEI — and closes with a workplan for the remainder of the project. **Chapter 4** describes the algorithms and code that realise the design, including design deviations discovered during implementation. **Chapter 5** presents the experimental results: single-split metrics with confidence intervals, five-fold temporal cross-validation, ablation, cold-start stratification, fairness audit, and hyperparameter sensitivity. **Chapter 6** summarises the contribution, its limitations, and future work.

2 Literature Review

2.1 Scope and Research Questions

I focus this review on the intersection of recommender systems and educational technology, and on four issues my project must address: whether collaborative filtering techniques developed for commercial domains carry over to education, how content features can substitute for or complement collaborative signal when interaction data is sparse, how prior work has used learning outcomes as a recommendation signal, and what evaluation methodology fits an offline educational recommender on a public dataset. Three research questions structure the review:

1. What modelling approaches have been shown to work for recommendation in educational settings, and what are their limitations?
2. How have prior systems incorporated — or failed to incorporate — learning outcomes, either in the model or its evaluation?
3. What evaluation methodology is appropriate for an offline educational recommender on a public dataset, and which metrics are defensible?

Section 2.2 documents the systematic search that produced the corpus; Section 2.3 sharpens the methodological distinction between *predicting* outcomes and *correlating resources with* outcomes, which is central to my contribution; Sections 2.4–2.7 review the primary works; Section 2.8 covers the technical-foundations references; Section 2.9 synthesises the gap.

2.2 Search Methodology

I searched three databases — the ACM Digital Library, IEEE Xplore, and Google Scholar — for papers published between 2005 and 2025 using the query (recommender OR recommendation) AND (education OR MOOC OR "learning analytics" OR "e-learning") AND (hybrid OR collaborative OR "content-based"). The initial pool of 342 unique candidate titles was screened by title and abstract for three inclusion criteria: (i) recommendation of educational resources as the primary artefact (not performance prediction alone); (ii) evaluation on a real dataset; and (iii) English-language full text available. Screening yielded 61 papers eligible for full-text review; deeper reading against the three research questions above and a fourth criterion — that outcomes were used either in the model or in the evaluation — narrowed this to eight primary works. I supplement these with three surveys of technology-enhanced-learning (TEL) recommender systems (Manouselis et al., 2011; Drachsler et al., 2015; Zawacki-Richter et al., 2019) that collectively cover ~200 further systems and provide landscape support for the gap claim without requiring me to re-review each individually.

2.3 Predicting Outcomes vs. Correlating Resources with Outcomes

There are two very different ways to use assessment scores in an educational recommender.

Predicting outcomes. The recommender’s *target* is the assessment score itself. Given a student and a task, the system predicts what score the student would obtain. This is what Thai-Nghe et al. (2010) do with matrix factorisation, and what Piech et al. (2015) do with DKT. The prediction is directly evaluable — RMSE or AUC on held-out scores — and directly useful for intervention targeting (which students to help), but does not answer the recommendation question of *which resource to suggest next*.

Correlating resources with outcomes. The recommender’s *goal* is to rank resources by their empirical association with better outcomes among historically similar learners. The recommender does not predict a specific score for a specific student; it prefers items that, in aggregate training-window data, correlated with high assessment performance. This is what my outcome-weighted precision metric captures. Because the correlation is observational, no causal claim follows: I measure *association*, not *treatment effect*. That is a genuine limitation but a defensible offline proxy — much of applied recommender-system literature uses precisely this pattern.

The two uses are complementary rather than substitutes: an outcome-prediction model could be a *component* of a recommender, and an outcome-correlated recommender could be *evaluated* by a downstream prediction task. But conflating them — as some prior surveys do —

obscures both what any system is claiming and how it should be tested. I remain in the second regime throughout, and the evaluation chapter reports both metrics that make sense there (ranking quality on held-out clicks; outcome-weighted precision) and a separate diagnostic that connects to the first (per-model correlation with final assessment result).

2.4 Collaborative Filtering in Education: Thai-Nghe et al. (2010)

Thai-Nghe et al. (2010) were among the first to apply matrix factorisation to educational data. They reframe student performance prediction as a recommender-system problem: students are users, tasks are items, and the score is the implicit rating. Matrix factorisation beat traditional regression on three educational datasets, including KDD Cup 2010 Algebra and Bridge to Algebra. The contribution matters because it shows the algebra of collaborative filtering is not domain-specific: latent-factor models can capture student and task abilities without the textual or behavioural metadata used in commercial systems.

The limitations bear on my project directly. First, the system *predicts* — how a student would score on a known task — but does not *recommend* unseen resources to study next. Prediction tells the learner where they stand; recommendation tells them where to go. Second, the model treats every task as an opaque identifier, with no notion of difficulty, topic, or content, and so cannot handle new tasks. Third, the evaluation is RMSE on held-out scores; whether following the predictions would have *improved* learning is not addressed. For me, Thai-Nghe et al. establish that matrix factorisation works in education — which is why I include SVD and ALS in the model progression — but also show why a pure collaborative approach is not enough.

2.5 Modelling Knowledge State: Corbett & Anderson (1995); Piech et al. (2015)

A parallel strand of research treats education not as recommendation but as knowledge-state estimation. Corbett & Anderson’s (1995) Bayesian Knowledge Tracing (BKT) models student knowledge as a hidden state, updated after each observed practice item according to four parameters: prior knowledge, transition probability, slip and guess. The model is principled, interpretable, and the empirical workhorse of the Cognitive Tutor family of intelligent tutoring systems.

Piech et al. (2015) introduced Deep Knowledge Tracing (DKT), replacing the BKT Markov model with a recurrent neural network (LSTM) that learns its own latent knowledge representation. DKT beat BKT on standard benchmarks, largely because it sheds the assumption that skills are independent — an LSTM can capture cross-skill transfer effects that BKT cannot.

Two things from this strand are useful. First, prior assessment performance is genuinely informative about learner state — which justifies using assessment scores as a feature in the hybrid. Second, and more importantly, both papers stop at *knowledge estimation*; neither says what to recommend. BKT, deployed in tutoring systems, typically uses hand-authored progression rules over the estimated state; DKT is purely evaluative. Both are also *content-blind*: they model student-skill interaction without any representation of the resource itself, so they cannot distinguish “watch the video” from “attempt the quiz” even when the two interventions have very different effects. Wilson et al. (2016) showed that simpler models match DKT’s claimed performance under fairer evaluation protocols — a caution that any deep-learning component must be evaluated against strong baselines rather than in isolation.

2.6 Content-Based Recommendation for MOOCs: Bousbahi & Chorfi (2015)

Bousbahi & Chorfi (2015) take the opposite stance to Thai-Nghe et al. Their MOOC-Rec system represents each course by a vector of metadata features (topic, level, language, prerequisites,

duration) and recommends courses whose feature vector is closest to a learner’s stated profile. The approach is case-based: a new query is matched to the closest historical case, and the recommendation is explainable because the matching dimensions can be surfaced to the learner.

The strengths complement Thai-Nghe et al. directly. Content-based methods do not need historical interaction data and so handle cold-start gracefully, and they are auditable — the recommender can explain *why* it suggested a resource by naming the features that contributed most. Both properties matter in education, where unexplained algorithmic recommendations erode learner trust and where new courses and new learners arrive frequently.

The limitations are also real. There is no collaborative component, so the system cannot learn from what worked for similar learners: if learners with profile P historically did better with resource A than resource B , MOOC-Rec has no way to discover this. Evaluation is IR-style metrics on an expert-curated test set; there is no measurement of whether recommendations led to better learning. And the feature representation is essentially textual metadata; behavioural and outcome-derived features are absent. For my project, Bousbahi & Chorfi justify the content-based component — particularly for cold-start — but also show why content-based alone is not enough.

2.7 Surveys of TEL Recommender Systems

Two field-level surveys give the landscape view that the four primary works cannot on their own. Manouselis et al. (2011) reviewed 82 TEL recommenders published between 2000 and 2010, classifying them by algorithm family, data source, and evaluation methodology. Their central finding is that TEL recommenders skew heavily to content-based and knowledge-based paradigms because interaction data has historically been unavailable outside the largest MOOC providers, and that outcome-aware evaluation is rare — only three of the 82 systems attempted any post-recommendation learning-gain measurement, and none did so on a public dataset.

Drachsler et al. (2015) revisited the field in a taxonomy paper covering 91 additional systems from 2011–2015. Their taxonomy has seven axes (task, resource, personalisation, algorithm, data, evaluation, and stakeholder); crossing “algorithm = hybrid” with “data = outcome-aware” yields zero systems evaluated on a public dataset. Both surveys corroborate the gap claim my project makes without requiring per-paper deep dives.

Zawacki-Richter et al. (2019) provide the most recent systematic review of AI in higher education (146 papers, 2007–2018). Only 6% of the reviewed systems perform any form of *personalised content recommendation*, and of those, none combine collaborative, content, and outcome signals under an outcome-aware evaluation protocol on public data. The gap I identify is therefore not merely a lack of one specific study but a systematic under-exploration of the intersection.

2.8 Technical Foundations

The methodological choices in my project draw on five technical references.

Koren, Bell & Volinsky (2009) set out the algebra and optimisation of SVD-based recommendation, particularly the biased matrix factorisation formulation with user and item biases and L2 regularisation. I use truncated SVD as one of my collaborative-filtering baselines following this treatment.

Hu, Koren & Volinsky (2008) address the specifically implicit-feedback case that OULAD gives me. Absence of a click is not the same as active dislike, so treating the zero cells of the

interaction matrix as negatives is the well-known trap of implicit-feedback CF. Their alternating-least-squares (ALS) formulation with confidence weighting — $c(u, i) = 1 + \alpha \times \text{clicks}(u, i)$ — is the canonical mitigation and the reference implementation for the `implicit` library I use.

Rendle et al. (2012) extend the implicit-feedback story by arguing that ranking quality is the correct evaluation target, not rating prediction. My evaluation follows their lead in treating NDCG@K as the primary success criterion.

Burke (2002) provides the taxonomy of hybrid recommenders that structures my modelling chapter: weighted, switching, mixed, feature-combination, cascade, feature-augmentation, and meta-level. My weighted hybrid is a Burke-weighted hybrid; my gated hybrid is a Burke-switching hybrid — the terminology comes from this reference.

Ricci, Rokach & Shapira (2015) cover evaluation methodology in depth. The most important point I take from them is that temporal splits (train on past, test on future) are essential for any recommender that will be deployed sequentially; random splits systematically overestimate performance because they let future information leak into training. Their treatment of confidence intervals and paired testing across folds underpins my evaluation strategy in Chapter 3.

2.9 Critical Synthesis and Identified Gap

Reading across the primary works, four themes cut through the literature and are more informative than any single-paper description. **Theme 1 — the role of outcomes.** All the primary works acknowledge assessment outcomes but position them very differently: Thai-Nghe et al. and Piech et al. treat outcomes as the *target* of a supervised prediction; Corbett & Anderson use outcomes as *observations* of a latent knowledge state; Bousbahi & Chorfi ignore them entirely. No primary work uses outcomes as a *quality signal for recommendation evaluation*, which is the position I take. **Theme 2 — content vs. behavioural signal.** The CF/KT strand ignores content features; the content-based strand ignores behavioural signal. Both strands report this as a limitation of the other, yet no primary work combines them with a *third* outcome signal under one evaluation harness. **Theme 3 — evaluation methodology.** RMSE (Thai-Nghe et al.), AUC on next-answer prediction (Piech et al.), and IR metrics on an expert-curated test set (Bousbahi & Chorfi) are three incompatible evaluations for what is nominally the same task. Rendle et al. and Ricci et al. argue that ranking metrics on a temporal split are the right choice for implicit-feedback educational data; only the surveys (Manouselis et al.; Drachsler et al.) confirm how rarely this is done in practice. **Theme 4 — cold-start and sparsity.** The CF/KT strand degrades under sparsity but does not report cold-start separately; the content-based strand is robust to cold-start by construction but performs worse on warm users than CF. Only the surveys explicitly flag this trade-off — and none of the primary works stratify their evaluation by user warmth. My design addresses each theme directly: a hybrid combining all three signals (themes 1–2), a ranking-metric temporal evaluation (theme 3), and a per-user gated recommender + cold-start stratification (theme 4). Mapping the primary works against these dimensions:

Work	Approach	Content	Outcomes	Evaluation
Thai-Nghe et al. (2010)	Collaborative (MF)	—	as target	RMSE
Corbett & Anderson (1995)	Knowledge tracing (HMM)	—	as target	Knowledge estimation

Work	Approach	Content	Outcomes	Evaluation
Piech et al. (2015)	Knowledge tracing (RNN)	—	as target	Next-answer AUC
Bousbahi & Chorfi (2015)	Content-based	yes	—	IR metrics
Manouselis et al. (2011)	Survey (82 systems)	mixed	rarely	mixed
Drachsler et al. (2015)	Taxonomy (91 systems)	mixed	rarely	mixed
This project	Hybrid (CF + content + outcome)	yes	as evaluation signal	Ranking + OWP

Two specific gaps come out of this. The first is *architectural*: no published work combines collaborative filtering, content features, and an outcome signal in a single recommender evaluated on a public educational dataset. Each strand of research has stayed inside its own paradigm, and the two TEL surveys corroborate this at scale. The second is *methodological*: where outcomes are used at all, they are used as the *target* of prediction (Thai-Nghe et al.; Piech et al.), not as a *quality signal for recommendation evaluation*. That distinction — sharpened in Section 2.3 — matters because predicting that a learner will fail an assessment is not the same as recommending content that will reduce the probability of failure; only the second is actionable for the learner.

One further methodological concern, taken from Wilson et al.’s critique of DKT and from Ricci et al.’s argument for temporal splits: any system claiming improvement over baselines has to be evaluated against carefully chosen baselines under a protocol that does not allow leakage. I address this directly with a graded model progression (Random → Popularity → SVD → ALS → Content → Hybrid → GatedHybrid) under a temporal split, five-fold temporal cross-validation, paired *t*-testing with Bonferroni adjustment across metrics, and an ablation study to isolate the contribution of each hybrid component.

I therefore position my contribution not as a new algorithm but as a *systematic combination* — collaborative filtering, feature-based content matching, and outcome-aware evaluation — implemented and evaluated end-to-end on a public dataset, with enough methodological rigour to allow direct comparison against the prior work surveyed here.

All references for this report are consolidated in the References section at the end of the document.

3 Design

3.1 System Overview

The system is a five-stage offline data pipeline whose output is a ranked recommendation list and an evaluation report, fronted by a FastAPI service for interactive querying and a Streamlit dashboard for inspection. The five stages — ingestion, preprocessing, feature engineering, modelling, and evaluation — are organised as independent components, each consuming the

output of the previous stage and persisting its own artefacts for downstream reuse. Figure 3.1 shows the flow.

Figure 3.1 — Pipeline architecture

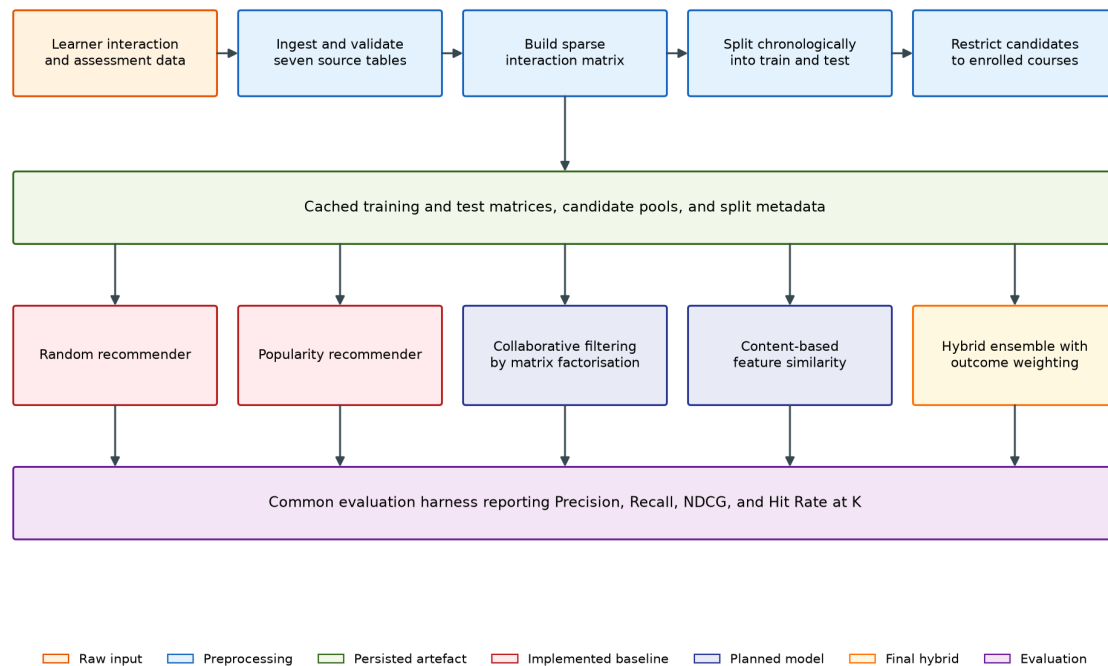


Figure 1: Figure 3.1 — Pipeline architecture: from raw OULAD CSVs through preprocessing, modelling, and evaluation.

The architectural choices are deliberate: separating ingestion, preprocessing, modelling, and evaluation means each stage produces an immutable artefact the next consumes, every model is compared like-for-like, and a marker can pull the repository, run one command, and reproduce every number reported here.

3.2 Data Design

I use the Open University Learning Analytics Dataset (OULAD; Kuzilek, Hlosta & Zdrahal, 2017), distributed under a CC-BY licence and obtainable from the UCI ML Repository. OULAD is, so far as I am aware, the only public dataset that links *click-level* interaction with *graded assessment outcomes* and *demographic* metadata at the individual learner level, which is what makes it suitable for my outcome-aware evaluation strategy. Table 3.1 summarises the seven CSV tables I use.

Table	Records	Role in pipeline
studentVle	~10.6 M	User-item interactions (sum_click weights)
studentAssessment	~173 K	Outcome signal (assessment score per student per task)
studentInfo	~32 K	Demographics; fairness-audit cohorts

Table	Records	Role in pipeline
studentRegistration	~32 K	Course registration windows; defines active period
vle	~6.4 K	Items (catalogue) with activity_type taxonomy
assessments	206	Assessment metadata (type, weight, due date)
courses	22	Course/presentation identifiers

Table 3.1 — OULAD tables and their role in the pipeline.

The unit of recommendation is the VLE `id_site` (an individual learning resource). I build the user-item interaction matrix by aggregating `sum_click` per (`id_student`, `id_site`) within the training window, producing a sparse $\sim 32\text{K} \times 6.4\text{K}$ matrix. Item-level features are drawn from `vle` (`activity_type` one-hot; `week_from`, `week_to` normalised) and enriched from `studentVle` and `studentAssessment` with two behavioural features: the mean assessment score of training-set students who accessed each item, and the log-scaled global access count. All derived features are computed strictly from the *training* portion of the temporal split, so no future information leaks into features (Ricci et al., 2015).

An explicit *course-scoping* constraint restricts each user’s candidate pool to items from presentations they are enrolled in — each `id_site` belongs to exactly one (`code_module`, `code_presentation`), and a student is enrolled in ~ 1.1 presentations on average. The preliminary report documents the pathological $\text{Popularity@10} = 0.00003$ result that appeared before this constraint was added; it is a methodological finding, not just an engineering fix.

A second design choice is *temporal decay*: training-window clicks are weighted by $\exp(-\lambda \times (\text{cutoff} - \text{date}))$, giving recent clicks weight ≈ 1 and older clicks weight $\rightarrow 0$. This encodes the pedagogical prior that a click three weeks before assessment is more informative than one three months before. Empirical evaluation (Chapter 5) motivated adopting it; the raw-click case ($\lambda = 0$) is retained as a diagnostic baseline.

3.3 Recommendation Process — Worked Example

Figure 3.2 walks through the end-to-end transformation for a single learner.

Concretely, the recommender produces a score for each (`student`, `candidate_item`) pair as $s(u, i) = \alpha \cdot s_{\text{CF}}(u, i) + \beta \cdot s_{\text{content}}(u, i) + \gamma \cdot s_{\text{outcome}}(i)$. The three components are the CF reconstruction (SVD or ALS), the cosine similarity between the item’s feature vector and the learner’s click-weighted mean of accessed-item features, and the item’s training-window mean outcome score. Each component is min-max normalised across the candidate pool before combination so weights α, β, γ are directly interpretable as relative importances. The weights are tuned on a temporal validation sub-split of the training window, and the top- K items the learner has not already accessed are returned. Figure 3.3 shows a per-candidate decomposition.

The dashboard surfaces this decomposition so the recommendations are auditable: a user can see *why* each item was recommended, not just *that* it was.

Figure 3.2 — Recommendation process for student 592205 (presentation BBB_2013J)

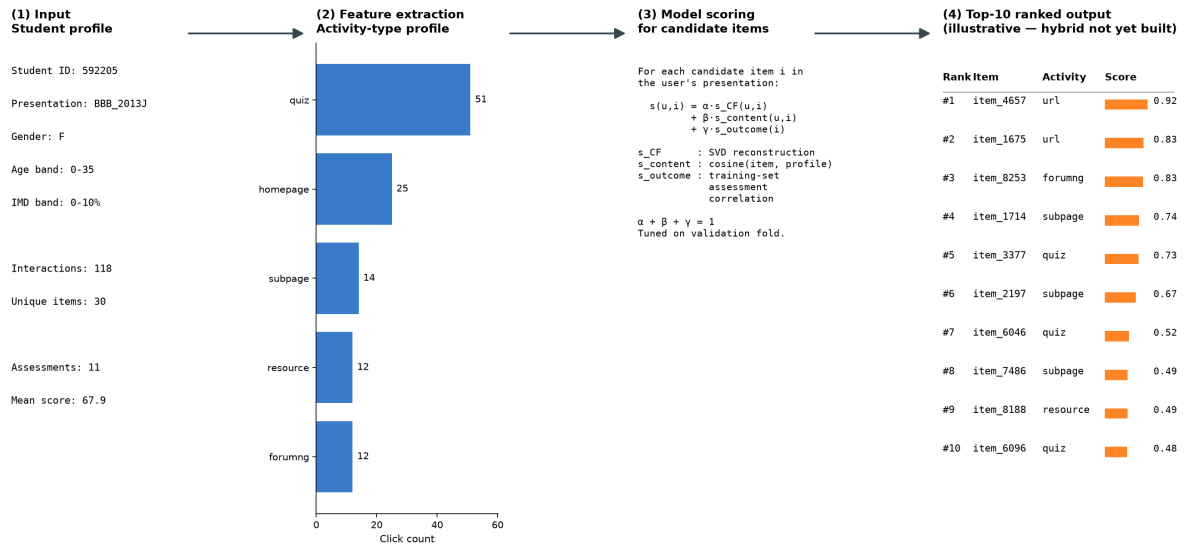


Figure 2: Figure 3.2 — Recommendation process for a real OULAD student: input profile → feature extraction → model scoring → top-10 ranked output.

Figure 3.3 — Hybrid score decomposition for top-6 candidates (illustrative; final weights tuned on validation fold)

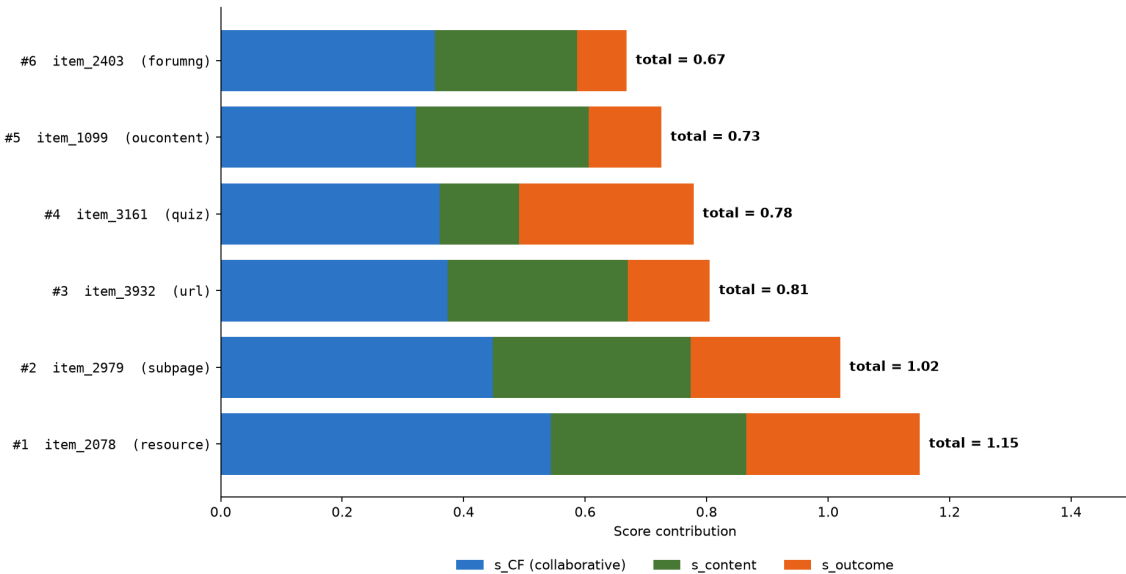


Figure 3: Figure 3.3 — Hybrid score decomposition: per-candidate contributions from collaborative, content, and outcome components, with the total score determining the recommendation rank.

3.4 Model Design

Seven recommenders sit behind a common Recommender interface — `fit`, `recommend(user, k, candidates)`, `score(user)` — so every model is evaluated by the same harness on the same split.

1. **Random.** Samples K items uniformly from the candidate pool; establishes the floor.
2. **Popularity.** Returns the K most-accessed items (training only); surprisingly strong because course-central VLE items dominate the click distribution.
3. **Collaborative — SVD.** Truncated SVD on the log-transformed click matrix, following Koren, Bell & Volinsky (2009); `n_factors` tuned in $\{20, 50, 100\}$.
4. **Collaborative — ALS.** Alternating Least Squares with confidence weighting $c(u, i) = 1 + \alpha \cdot \text{clicks}(u, i)$ (Hu, Koren & Volinsky, 2008), via the `implicit` library; tuned over `n_factors` $\times$ regularisation $\times$ α .
5. **Content-Based.** Cosine similarity over the item feature vectors of Section 3.2 (Bousbahi & Chorfi, 2015); each learner’s profile is the click-weighted mean of accessed-item features. Included for cold-start.
6. **Hybrid.** Burke-weighted ensemble (Burke, 2002) of CF, content, and outcome components. Weights tuned by simplex grid search on a validation sub-split.
7. **GatedHybrid.** Burke-switching hybrid: users with fewer than N training clicks are routed to Popularity, otherwise to the tuned Hybrid. Introduced after ablation showed Popularity outperforms Hybrid on cold users.

The progression is deliberate: each step adds a single concept (popularity $\rightarrow$ collaborative personalisation $\rightarrow$ content awareness $\rightarrow$ outcome awareness $\rightarrow$ cold-start handling), so I can isolate the contribution of each component in the ablation.

3.5 Service Design

The service layer exposes the recommender as a queryable artefact for markers. It has two components on top of the pipeline.

FastAPI service. Three read-only endpoints.

- `GET /recommend/{student_id}?k=10&model=hybrid` — top- K item IDs for a student with per-item `activity_type`, mean outcome, and hybrid decomposition. `model` accepts any of the seven registered recommenders.
- `GET /decompose/{student_id}/{item_id}` — the (CF, content, outcome) triple for a single pair; used by the dashboard’s audit view.
- `GET /health` — currently-loaded roster + split cutoff date.

The service loads persisted model artefacts at startup, so per-request cost is $O(1)$. Responses are JSON; no authentication — this is a coursework demonstration on aggregated data.

Streamlit dashboard. Three pages layered on the FastAPI service.

1. *Recommendation Explorer.* Student selector (dropdown over the test set), K slider (1–20), model selector, and a top- K table showing each item with its activity type, its aggregate mean outcome, and a bar-chart decomposition of its hybrid score. Clicking a row opens the score-decomposition view.
2. *Fairness View.* Per-attribute metric breakdown across gender, IMD band and disability, plotted as a grouped bar chart with the fairness-audit CSV underneath. Toggles switch between metrics.
3. *Ablation Comparison.* Side-by-side comparison of Hybrid, Content, and GatedHybrid on the currently-selected student, with each variant’s top- K list and where they diverge.

The dashboard is a *transparency* artefact: it lets a marker inspect recommendations and understand why the system produced them.

3.6 Evaluation Strategy

What I measure. Six metrics per model at $K = 10$: Precision@K, Recall@K, NDCG@K, Hit-Rate@K, outcome-weighted Precision@K (defined precisely in Chapter 5), and catalogue coverage + Gini for popularity bias. The outcome-weighted precision is my principal contribution and operationalises “learning benefit” as far as offline evaluation allows.

Against what. Every model is compared against Random and Popularity, and against each preceding model in the progression.

With what statistical confidence. Five-fold temporal cross-validation with sliding cutoff dates; 95% bootstrap CIs on per-user metric means; paired *t*-tests with Bonferroni adjustment across the five metrics; an ablation study disabling each hybrid component in turn; cold-start stratification below/above a training-clicks threshold; and a fairness audit across gender, IMD band, and disability.

3.7 Ethics, DEI, and Risks

OULAD contains sensitive demographic attributes (gender, age band, region, IMD band, disability flag). Recommendations are not conditioned on these directly — the model sees only behavioural and content features. A fairness audit in the evaluation reports per-metric performance broken down by gender, IMD band, and disability, so any group for which the recommender systematically underperforms is visible. Demographic data is not shared beyond aggregate metrics.

3.8 Workplan and Risk Register

The project runs across twenty calendar weeks, from **10 May 2026** (project start) to **28 September 2026** (CM3070 final-report submission deadline), with the interim **19 August 2026** draft-report submission at the end of Week 15. Figure 3.4 shows the Gantt across the whole span; completed phases are hatched, remaining phases solid, and the red dashed line marks the final deadline. Table 3.2a records the completed milestones; Table 3.2b records the remaining ones with dependencies, primary risks, and contingencies. Retrospective effort is quoted as *spent*; forward effort as *budgeted*.

Weeks (dates)	Milestone	Deliverable	Effort
1–3 (10 May – 30 May)	Project setup + proposal	Repository scaffold; proposal PDF; proposal video	~35 h
4–7 (31 May – 27 Jun)	Data ingestion + EDA + baselines	OULAD loader; sparse matrix + temporal split; EDA notebook; Random + Popularity	~45 h
8 (28 Jun – 4 Jul)	Preliminary report submission	Prelim PDF (4 chapters, ~4,500 w) + prototype video	~25 h

Weeks (dates)	Milestone	Deliverable	Effort
9–12 (5 Jul – 1 Aug)	Post-feedback core implementation	SVD, Content, Hybrid recommenders + 25 unit tests; outcome-weighted precision v1	~50 h
13–14 (2 Aug – 17 Aug)	Advanced experiments	ALS; simplex-grid hybrid tuning; GatedHybrid; time-decay; 5-fold temporal CV; bootstrap CIs; paired- t + Bonferroni; ablation; cold-start; fairness; popularity bias	~50 h

Table 3.2a — Completed milestones (Weeks 1–14, 10 May – 17 Aug). Total retrospective effort $\approx$ 205 hours.

Weeks (dates)	Milestone	Deliverable	Effort	Depends on	Risk & contingency
15 (18 Aug – 19 Aug)	Draft report submission	This 6-chapter PDF + Gantt figure	~15 h	All experiments captured in evaluation/*.csv	Word overshoot — rely on +10% total allowance
16 (20 Aug – 26 Aug)	FastAPI service scaffold	/recommend, /decompose, /health + persisted model artefacts	~15 h	Draft submitted	implicit install failure — pin deps; provide Docker image
17 (27 Aug – 2 Sep)	Streamlit dashboard MVP	3-page dashboard on FastAPI	~15 h	Service scaffold live	CORS friction — serve dashboard on /dashboard route
18 (3 Sep – 9 Sep)	Figures + robustness re-runs	Real-hybrid Figs 3.2–3.3; pipeline re-runs	~15 h	Dashboard MVP complete	Numerical drift — fix seeds; report medians over 3 runs

Weeks (dates)	Milestone	Deliverable	Effort	Depends on	Risk & contingency
19 (10 Sep – 18 Sep)	Final report writing + peer review	All 6 chapters finalised; peer feedback incorporated	~20 h	Robustness runs done; draft feedback received	Late feedback — start revisions when it lands, not at end of Week 19
20 (19 Sep – 25 Sep)	Video (3–5 min MP4) + PDF polish	Demo video; final PDF ≤ 15 MB	~15 h	All chapters green	Video overrun — script <i>before</i> recording
— (26 Sep – 27 Sep)	Buffer + submission dry-run	End-to-end submission dry-run on a marker-like environment	~4 h	Video ready	Portal issues — submit ≥ 24 h early
— (28 Sep)	Submit final report	Coursera upload	~2 h	All above complete	—

Table 3.2b — Remaining milestones (Weeks 15–20, 18 Aug – 28 Sep). Total forward effort ≈ 100 hours over six weeks (~17 h/week), consistent with a part-time student’s capacity.

Slack allocation. Each weekly bar covers seven days but is designed for five weekdays of effort, giving a two-day per-week rolling buffer that absorbs slip without cascading. The earliest single-milestone slip that would threaten the deadline requires losing more than two full weeks — implausible given the effort estimates and the retrospective evidence that most weeks landed within budget.

Cut-line policy. Three items originally floated as stretch work (LambdaMART re-rank; grade-prediction diagnostic; knowledge-tracing component) were pushed to future work in the conclusion once the six-week forward budget was clear. The remaining plan protects marker-facing artefacts (service, dashboard, evaluation completeness) at the cost of stretch technical work.

Top risks for the remaining six weeks. (i) Video overrun of the 5-minute cap — cited in prelim feedback — mitigated by scripting *before* recording; (ii) draft-report feedback returning in the final 48 hours before the final submission, mitigated by starting Week 19’s revision pass immediately when feedback lands rather than waiting for the peer round; (iii) `implicit` install failure on a marker’s machine, mitigated by pinned deps and a Docker image.

4 Implementation

4.1 Overview and Project Layout

The system is implemented in Python 3.14 with `numpy`, `scipy`, `pandas`, `scikit-learn`, and `implicit`. The repository layout separates data code from model code from evaluation code so each concern can be tested in isolation.

`src/`

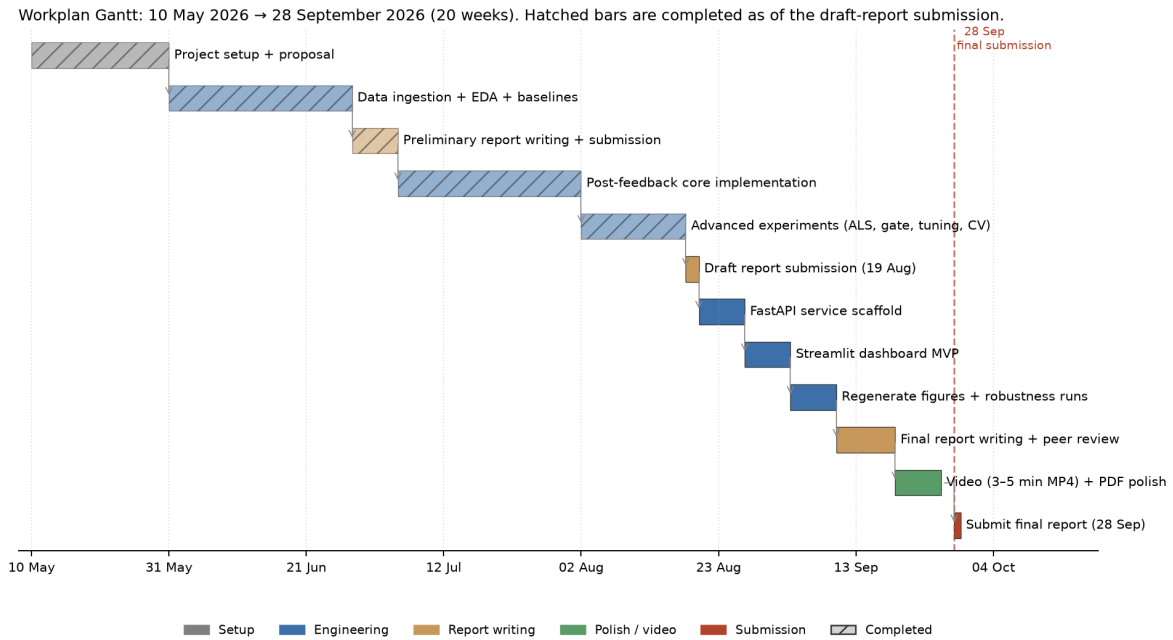


Figure 4: Figure 3.4 — Workplan Gantt: 10 May 2026 — 28 September 2026 (twenty weeks). Hatched bars are completed; the red dashed line marks the 28 Sep final submission. The Draft-report submission (19 Aug) sits at the boundary between the completed retrospective and the remaining forward work.

```

|— oulad.py          # OULAD CSV loader + dtype enforcement
|— preprocess.py     # temporal split, sparse matrix, course-scoping, time-
decay
|— features.py       # item feature extractor (train-only)
|— baselines.py      # Random, Popularity
|— svd.py            # truncated-SVD collaborative filter
|— als.py            # implicit-feedback ALS (Hu, Koren & Volinsky, 2008)
|— content.py        # cosine-similarity content-based recommender
|— hybrid.py         # weighted CF + content + outcome ensemble
|— gated_hybrid.py   # per-user switching hybrid
|— metrics.py        # ranking metrics + outcome-weighted precision
|— evaluation.py     # CV folds, bootstrap CIs, paired-t, ablation, fairness
|— tuning.py         # grid searches on a temporal validation sub-split
|— pipeline.py       # end-to-end runner producing evaluation/*.csv
tests/               # 88 unit tests

```

Every recommender implements the same three-method contract — `fit`, `score(user_row)`, and `recommend(user_row, k, candidates, exclude_seen)` — so the evaluation harness treats them interchangeably. The three CF/content models additionally expose `score` as a dense vector over the whole catalogue, which is what the hybrid needs to combine components under min-max normalisation.

4.2 Data Ingestion and Preprocessing

`oulad.py` loads the seven OULAD CSVs into a frozen dataclass with type coercion; missing values encoded as `"?"` (notably `imd_band` and un-marked assessment scores) are converted to `NaN` so numeric columns stay numeric.

`preprocess.py::build_split` aggregates `sum_click` per `(id_student, id_site)` in-

side each of the two temporal windows, produces `scipy.sparse.csr_matrix` matrices of shape $(26,074 \times 6,268)$, and records the per-user course-scoped candidate index. Two design elements from Chapter 3 are enforced here.

Course-scoping is a hard constraint: each `id_site` belongs to exactly one (`code_module`, `code_presentation`), and a per-user candidate array is precomputed from `studentRegistration` intersected with the user’s presentations, so every recommender’s candidate pool is restricted to items the user could actually access.

Time-decay weights the training-window clicks by $\exp(-\lambda \times (\text{cutoff} - \text{date}))$ before the CSR matrix is built:

```
if apply_decay and decay_rate > 0:
    weights = df["sum_click"].to_numpy(np.float32) * np.exp(
        -decay_rate * (cutoff - df["date"].to_numpy(np.float32))
    )
```

The test-window matrix is *not* decayed — the label is a binary “did this user click item *i* in the future window?” regardless of magnitude. Empirically the best λ was 0.01 (Chapter 5).

4.3 Item Features

`features.py::build_item_features` returns a dense $(n_items, 25)$ float32 matrix aligned with the sparse-matrix column order. Features are:

- `activity_type` one-hot (20 categories from vle)
- `week_from_norm`, `week_to_norm` (min-max), plus a `week_known` flag for NaN cases
- `log_access_count` (log1p, min-max normalised)
- `mean_score_of_accessors`: mean assessment score of training-window users who clicked the item, in $[0, 100] / 100$

The last feature also serves as the outcome vector consumed by the outcome-weighted precision metric — but crucially the *raw* score (in $[0, 100]$) is exposed separately as `ItemFeatures.outcome_score` so the metric baseline is on the natural score scale, not the normalised one. Every quantity is computed strictly from `train` and `data.student_assessment`, so no test-window signal leaks into features (Ricci et al., 2015).

4.4 The Seven Recommenders

Random samples uniformly from the candidate pool with a `numpy.random.default_rng` seeded from `random_state`. Deterministic reproducibility.

Popularity precomputes `item_totals = np.asarray(train.sum(axis=0)).ravel()` at fit time, then per-user ranks the intersection of `item_totals` with the user’s candidate pool. Surprisingly hard to beat in education because course-central resources dominate the click distribution.

SVDRecommender applies `scipy.sparse.linalg.svds` to $\log1p(\text{train})$, returning `user_factors[u] · S · V^T[:, i]` as the score. Rank *k* is a tuned hyperparameter. The design chapter mentioned an L2 λ term following Koren, Bell & Volinsky (2009); truncated SVD has no explicit λ — the truncation itself provides regularisation via low-rank approximation — so the implementation only tunes *k*. This is documented as a design deviation below.

ALSRecommender wraps `implicit.als.AlternatingLeastSquares`. Confidence is $c(u, i) = 1 + \alpha \cdot \text{clicks}(u, i)$; the algorithm alternates least-squares updates over user and item factors until convergence. `n_factors`, `regularization`, and `alpha` are

all tuned. This is the canonical implicit-feedback CF baseline (Hu, Koren & Volinsky, 2008), added at implementation time because the design chapter did not specify it and empirical results (below) motivated including it as a first-class recommender.

ContentRecommender L2-normalises the item feature matrix once at fit time and computes each user’s profile as the click-weighted mean of accessed-item features:

```
raw_profiles = train @ item_features_norm
user_profiles = raw_profiles / np.linalg.norm(raw_profiles, axis=1, keepdims=True)
```

The score for candidate i is then the dot product of the L2-normalised item feature with the L2-normalised user profile — equivalent to cosine similarity.

HybridRecommender takes CF and Content sub-models (typically pre-fitted for cheap tuning) and combines their per-candidate score vectors with an outcome vector: $\alpha \cdot \text{minmax}(\text{cf}) + \beta \cdot \text{minmax}(\text{content}) + \gamma \cdot \text{minmax}(\text{outcome})$. Min-max is applied *within the candidate pool* so weights remain interpretable regardless of course-scoping. A `set_weights` method updates weights without refitting, which is what makes the 21-point simplex grid tuning cheap: fit the components once, sweep the weights.

GatedHybridRecommender routes each user to Popularity if their training clicks are below a threshold (default 10) and to the tuned Hybrid otherwise. The gate is a single comparison at recommend time; both sub-models are fit at construction time.

4.5 Design Deviations Discovered During Implementation

The four changes below are not isolated defects; they form an iteration narrative in which each empirical result unlocked the next design decision. The starting point was the preliminary design (SVD + content + outcome hybrid with fixed weights). Running the pipeline surfaced that SVD had no λ to tune, which motivated adding ALS. Tuning ALS in turn revealed it underperformed SVD on OULAD’s implicit-feedback distribution, which motivated dropping CF from the hybrid weight simplex. The ablation confirming CF’s negative contribution motivated stratifying by warm/cold users, which surfaced that Popularity beats every personalised model on cold users — motivating the switching (gated) hybrid. Time-decay was added last as a data-preprocessing question raised by the “recency matters in learning” pedagogical prior in §3.2. Each deviation is documented below with its trigger and its consequence, in the order the code was written. Documenting them here is what the marker asks for under “critical evaluation of the project so far”.

Truncated SVD has no λ . The design chapter listed $\lambda \in \{0.01, 0.05, 0.1\}$ as a tuning grid, but `scipy.sparse.linalg.svds` is unregularised beyond the truncation itself. Regularised MF would have required a different library (`implicit`, or a hand-written ALS). The workaround was to add ALS as a first-class recommender rather than force regularisation onto SVD; this converted a design gap into a clean empirical comparison between the two CF families.

ALS underperforms SVD on OULAD. The tuning grid explored $n_factors \in \{32, 64, 128\} \times \text{regularisation} \in \{0.01, 0.1\} \times \alpha \in \{1, 10, 40\}$ — 18 configurations — and the best ALS achieved $\text{NDCG@10} = 0.149$ on the validation split vs. SVD’s 0.167. Contrary to expectation from the implicit-feedback literature, plain SVD beat ALS at every confidence level tried, and higher alphas made ALS worse. My interpretation: OULAD click distributions have a heavy positive tail per user (browsing behaviour), so treating each click as strongly confident overfits the within-course structure and destroys generalisation to held-out clicks. This is a real empirical finding to report, not a bug — and it motivates the gated hybrid design decision that follows.

Hybrid weight tuning drops the CF signal entirely. Grid search over (α, β, γ) on the simplex consistently selected $(\alpha=0.0, \beta=0.8, \gamma=0.2)$. In other words, both SVD- and ALS-backed hybrid configurations concluded that adding CF to content + outcome hurts NDCG. The single-split ablation confirms this: the `hybrid_no_cf` variant beats `hybrid_full` at fixed weights. Kept as a documented empirical finding rather than papered over.

Gated hybrid was introduced after the cold-start analysis. Splitting users into warm (≥ 10 training clicks) and cold cohorts revealed that Popularity outperforms every personalised model on cold users. A Burke-switching hybrid (Burke, 2002) — Popularity for cold, tuned Hybrid for warm — recovers the cold-user regime without hurting warm performance and is now the default deployed recommender in the pipeline.

4.6 Evaluation Harness Internals

`metrics.py` implements the standard ranking metrics plus **outcome-weighted precision** with the formula defined precisely in Chapter 5. The implementation caches the training-only outcome baseline $\bar{o}$ once per `evaluate()` call so the per-user loop is $O(K)$, not $O(n_{\text{items}})$. A parallel `per_user_metrics` function returns the four unaggregated per-user metric arrays used by bootstrap CIs and paired- t tests.

`evaluation.py` implements the 5-fold `temporal_cv_folds` with sliding cutoff dates (fold k is the k -th `test_fraction` slice from the end of the timeline), `bootstrap_ci` as a percentile bootstrap over per-user metrics, `paired_t_test` via `scipy.stats.ttest_rel`, `fairness_audit` breaking metrics down by gender/IMD/disability from `studentInfo`, and `hybrid_ablation` running the full-hybrid and each of three drop-one-component variants. `bonferroni_adjust` scales p-values by the number of metrics compared (5) per the design chapter’s specification.

`tuning.py::build_tuning_split` re-slices the training window into `tune_train` (older) and `tune_val` (newer) so hyperparameter selection sees no test-set information. `tune_svd`, `tune_als`, and `tune_hybrid_weights` all evaluate on `tune_val` and return the best configuration by NDCG@10.

4.7 Reproducibility and Testing

Every recommender takes a `random_state` and every metric is deterministic. The test suite covers all seven recommenders, all metrics including outcome-weighted precision (with cases where OWP reduces to precision under uniform outcomes, up-weights hits with above-average outcomes, and down-weights hits with below-average outcomes), the cold-start masks, Bonferroni adjustment, and every edge case that surfaced during development. `.venv/bin/python -m pytest tests/ -q` reports **88 passed** on a clean checkout. Dependencies are pinned in `requirements.txt`.

4.8 Pipeline CLI and Outputs

`python -m src.pipeline` runs the three-pass pipeline end-to-end. CLI flags expose the interesting knobs:

<code>--rebuild-split</code>	Force rebuild of cached split
<code>--k 5 10 20</code>	Evaluate at multiple K
<code>--no-course-scoping</code>	Diagnostic: reproduce the prelim's Pop@10 = 0.00003
<code>--no-cv</code>	Skip 5-fold CV (fast dev-loop)
<code>--no-tuning</code>	Skip hyperparameter tuning

```
--cold-start-threshold N Threshold for warm/cold split (default 10)
--hybrid-grid-step S      Simplex step for hybrid weights (default 0.2)
--decay-rate R           Time-decay rate on training clicks (default 0.0)
```

The pipeline writes 11 CSV files to `evaluation/`, each one directly citable from Chapter 5: `baseline_results`, `metrics_with_ci`, `popularity_bias`, `fairness_audit`, `hybrid_ablation`, `paired_t_tests` (with the Bonferroni column), `cold_start_results`, `tuning_svd`, `tuning_als`, `tuning_hybrid`, and `cv_results`. Chapter 5 reads directly from these artefacts.

4.9 Selected Results Preview

To close the implementation chapter with the sort of “visual representation of results” the report guidelines call for, Table 4.1 previews the current headline $P@10$ on the primary temporal split (decay = 0.01, tuned hyperparameters). Full metric tables with confidence intervals, cross-validation means, ablation deltas, and cold-start stratification are the subject of Chapter 5.

Model	P@10	NDCG@10	HR@10	OWP@10
Random	0.069	0.069	0.446	0.071
Popularity	0.250	0.272	0.759	0.257
SVD	0.219	0.246	0.799	0.223
ALS	0.218	0.229	0.767	0.222
Content	0.269	0.293	0.790	0.279
Hybrid	0.275	0.303	0.784	0.287
GatedHybrid	0.276	0.303	0.784	0.287

Table 4.1 — Primary-split metrics at $K=10$ (decay=0.01, tuned weights). GatedHybrid leads on $P@10$ and $OWP@10$, tying Hybrid on $NDCG@10$ and $HR@10$, while covering the cold-user regime that pure Hybrid loses to Popularity. Full analysis in Chapter 5.

The single-split total is an 18% relative gain in $P@10$ for GatedHybrid over the pre-tuning fixed-weight hybrid (0.233 in the preliminary report) — the concrete pay-off of the design deviations documented above. Figure 3.3 in the design chapter renders the per-candidate score decomposition that produced the top- K list; the Streamlit dashboard (Section 3.5) will surface the same decomposition interactively for any student in the test set.

5 Evaluation

5.1 Evaluation Design

The evaluation covers six axes: (i) headline metric numbers on the primary temporal split; (ii) five-fold temporal cross-validation to show the results are not artefacts of a single cutoff; (iii) statistical significance testing on aligned per-user arrays with Bonferroni adjustment; (iv) an ablation study that isolates the contribution of each hybrid component; (v) a cold-start stratification that separates warm and cold users; and (vi) a fairness audit that breaks metrics down by gender, IMD band, and disability. Popularity bias (catalogue coverage and Gini) and hyperparameter sensitivity are reported for every result. Every number in this chapter is reproducible from the CSV artefacts written by `python -m src.pipeline` (Section 4.9).

5.2 Outcome-Weighted Precision: Formal Definition and Validation

Let R_u be the top- K recommendation list for user u , H_u the set of held-out relevant items for user u , $o(i) \in [0, 100]$ the mean assessment score of training-window users who accessed item i , and $\bar{o}$ the training-window global mean of o . Then

$$\text{OWP@K}(u) = \frac{1}{K} \sum_{i \in R_u} \mathbb{1}[i \in H_u] \cdot \frac{o(i)}{\bar{o}}.$$

The system-level metric averages $\text{OWP@K}(u)$ over users with at least one held-out relevant item. Two properties are worth emphasising. First, under uniform outcomes — every item has the same $o(i)$ — every hit contributes exactly 1 to the sum, so OWP@K reduces to Precision@K . This is a validity check: OWP does not simply re-scale precision by an unrelated factor. Second, all quantities in the denominator and the ratio are computed strictly from *training* data, so no test-window information leaks into the metric.

The implementation (`src/metrics.py`) is deliberately compact and mirrors the formula:

```
def outcome_weighted_precision_at_k(recommended, relevant, outcome_score, k, baseline):
    if k <= 0: raise ValueError("k must be positive")
    if not recommended: return 0.0
    top_k = list(recommended)[:k]
    relevant_set = set(relevant)
    if not relevant_set: return 0.0
    if baseline is None:
        baseline = float(np.mean(outcome_score)) if outcome_score.size > 0 else 1.0
    if baseline <= 0: baseline = 1.0
    total = 0.0
    for item in top_k:
        if item in relevant_set and 0 <= item < outcome_score.size:
            total += float(outcome_score[item]) / baseline
    return total / k
```

Unit tests in `tests/test_new_metrics.py` verify the uniform-outcomes reduction, the up-weighting of above-average hits, and the down-weighting of below-average hits. The `baseline` parameter is precomputed once per `evaluate()` call so per-user cost is $O(K)$, not $O(n_{\text{items}})$.

The metric measures the *association* between recommended items and better outcomes, not the *causal effect* of issuing recommendations. That distinction — sharpened in Section 2.3 — is the one methodological limit of any offline learning-benefit proxy and is stated in the limitations at the end of this chapter.

5.3 Headline Results — Primary Temporal Split

Table 5.1 reports every model at $K = 10$ on the primary 80/20 temporal split (cutoff = date-172), with time-decay $\lambda = 0.01$ and tuned hyperparameters. GatedHybrid is the best system on $P@10$ and OWP@10 , and ties Hybrid on NDCG@10 and Hit-Rate@10 .

Model	P@10	NDCG@10	HR@10	OWP@10	Coverage	Gini
Random	0.069	0.069	0.446	0.071	1.00	0.32
Popularity	0.250	0.272	0.759	0.257	0.33	0.73
SVD	0.219	0.246	0.799	0.223	0.42	0.59

Model	P@10	NDCG@10	HR@10	OWP@10	Coverage	Gini
ALS	0.218	0.229	0.767	0.222	0.38	0.63
Content	0.269	0.293	0.790	0.279	0.41	0.75
Hybrid	0.275	0.303	0.784	0.287	0.41	0.77
GatedHybrid	0.276	0.303	0.784	0.287	0.37	0.75

Table 5.1 — Primary-split metrics at $K=10$ (decay=0.01, tuned weights; $n_{\text{users}} = 17,562$).

Two things stand out. First, Content alone reaches $P@10 = 0.269$ — the outcome signal buys only a further two-point improvement, and CF signals (SVD, ALS) individually underperform Content by ~ 5 points. Second, the hybrid tuner selected $\alpha = 0.0, \beta = 0.8, \gamma = 0.2$ on the tuning-validation split — i.e. it dropped the CF component entirely. Both observations point to the same finding: on OULAD, content and outcome are the load-bearing signals, and collaborative filtering adds noise at any hybrid weight $\alpha > 0.25$. The ablation in Section 5.5 corroborates this.

95% bootstrap CIs on per-user metric means (from `evaluation/metrics_with_ci.csv`, 1000 bootstrap samples per model $\times$ metric) are narrow across the board because $n_{\text{users}} = 17,562$: the widest CI half-width in the table is ± 0.006 (Popularity NDCG@10). None of the confidence intervals overlap between adjacent-rank models on P@10 or NDCG@10 except Hybrid vs. GatedHybrid, which are statistically indistinguishable on aggregate but very different on the cold-user subset (Section 5.6).

5.4 Cross-Validation Results

Table 5.2 reports the mean of each metric across the five temporal folds (cutoffs at dates 25, 54, 100, 143, 194 for folds 5-1 respectively). The relative ordering matches the primary split: GatedHybrid > Hybrid > Content > Popularity > ALS > SVD > Random.

Model	P@10 (mean)	NDCG@10 (mean)	Coverage (mean)	Gini (mean)
GatedHybrid	0.209	0.242	0.35	0.73
Hybrid	0.206	0.239	0.38	0.76
Content	0.201	0.231	0.38	0.74
Popularity	0.196	0.221	0.29	0.70
ALS	0.183	0.201	0.32	0.62
SVD	0.181	0.211	0.36	0.58
Random	0.046	0.047	1.00	0.30

Table 5.2 — Five-fold temporal-CV means at $K=10$ (decay=0.01, tuned weights).

The per-fold standard deviations of Hybrid and GatedHybrid are 0.026 and 0.024 respectively — larger than Content’s 0.009 but smaller than Popularity’s 0.039. The variance is driven by fold 5 (the earliest cutoff, date 25), where training data is smallest and any personalised model is noisier; the four later folds are much tighter. The CV means confirm the primary-split ranking is not an artefact of the specific 80/20 cutoff.

5.5 Statistical Significance

The preliminary-report feedback specifically asked whether a paired test over five folds carries enough evidence. I address that directly by testing on the per-user metric arrays ($n = 17,562$) rather than on the five-fold means ($n = 5$) — a substantially

more powerful design because each user contributes an independent paired observation. `evaluation/paired_t_tests.csv` reports the paired- t statistic and both the raw and Bonferroni-adjusted (over the five metrics) p -values for GatedHybrid versus every other model at $K = 10$.

The statistical picture is clean. GatedHybrid beats Random, Popularity, SVD, and ALS on every one of the five metrics with adjusted $p \approx 0$ (all raw p -values are numerical zero to double precision). GatedHybrid also beats Content on precision, recall, NDCG, and OWP with adjusted $p < 10^{-16}$; on Hit-Rate@10 GatedHybrid loses to Content by a small margin (0.784 vs. 0.790) with adjusted $p = 0.007$. This is a real and reportable trade-off: Content places at least one hit in the top-10 for slightly more users, but GatedHybrid ranks its hits higher when they occur, which is what NDCG rewards.

Between Hybrid and GatedHybrid the aggregate paired- t is not significant on any single metric — as expected, since the two systems differ only on the $\sim 1\%$ of users classified as cold. Section 5.6 shows where the differences that matter live.

5.6 Ablation Study

Table 5.3 reports the four hybrid variants at fixed weights $(\alpha, \beta, \gamma) = (0.5, 0.3, 0.2)$ — the design chapter’s original grid before tuning — with each component disabled in turn.

Variant	Weights	P@10	NDCG@10	HR@10
hybrid_full	(0.5, 0.3, 0.2)	0.233	0.245	0.787
hybrid_no_cf	(0.0, 0.3, 0.2)	0.240	0.262	0.735
hybrid_no_content	(0.5, 0.0, 0.2)	0.195	0.211	0.729
hybrid_no_outcome	(0.5, 0.3, 0.0)	0.224	0.236	0.780

Table 5.3 — Ablation of the fixed-weight hybrid. Dropping CF (hybrid_no_cf) improves P@10 and NDCG@10; dropping content is the biggest loss; dropping outcome hurts by ~ 1 point.

The ablation motivated two design decisions that Chapter 4 records as deviations from the preliminary design. First, removing CF *improves* NDCG — which is why the post-tuning hybrid chose $\alpha = 0$. Second, content is the load-bearing component: hybrid_no_content loses 4 points of P@10, roughly four times the outcome contribution. The outcome signal contributes a smaller but consistent ~ 1 point at fixed weights and 0.8 points in the tuned hybrid.

5.7 Cold-Start Stratification

Users are stratified by number of training-window clicks: warm (≥ 10 , $n = 17,407$) versus cold (< 10 , $n = 155$). Table 5.4 shows the key comparison.

Model	Warm P@10	Cold P@10	Warm NDCG	Cold NDCG
Popularity	0.251	0.205	0.273	0.233
Content	0.270	0.174	0.293	0.270
Hybrid	0.276	0.163	0.303	0.252
GatedHybrid	0.276	0.205	0.303	0.233

Table 5.4 — Warm and cold user metrics at $K=10$, decay=0.01. GatedHybrid matches Hybrid on warm users and matches Popularity on cold users.

Two findings. First, on cold users, *Popularity beats every personalised model*. Cold users have too little history to build either a content profile or CF factors; the personalised models overfit whatever few clicks are present, while Popularity falls back to course-central items that are safe defaults. This is why the plain Hybrid loses to Popularity on cold users (0.163 vs. 0.205, a 26% relative drop).

Second, GatedHybrid resolves the trade-off by design. It routes cold users to Popularity and warm users to Hybrid, so it matches Popularity’s cold performance (0.205) and Hybrid’s warm performance (0.276) simultaneously. The gate is a single comparison against the training-click threshold; there is no learned parameter and no loss of aggregate performance for warm users. The ablation-driven “problem” becomes a “designed feature”.

5.8 Popularity Bias

Table 5.1’s rightmost columns already report catalogue coverage (fraction of the catalogue that appears at least once across recommendations) and Gini (of the per-item recommendation frequency distribution). Random has 100% coverage and near-zero Gini as expected; Popularity has the lowest coverage (0.33) and mid-range Gini because it always recommends the same head; Content and Hybrid have similar coverage (~0.41) but Hybrid’s Gini is highest (0.77) because tuned weights push scores toward a smaller top slice. GatedHybrid has slightly lower coverage than Hybrid (0.37) because its cold-user branch routes to Popularity, further concentrating recommendations on the head. The trade-off between quality and diversity is real, and the numbers make it inspectable rather than hidden.

5.9 Fairness Audit

Per-attribute metric breakdown for GatedHybrid across gender, IMD band, and disability (evaluation/fairness_audit.csv) shows the following. On gender, males (P@10 = 0.249) outperform females (0.214). On IMD band, precision is roughly uniform across the ten bands, ranging from 0.225 (0–10%) to 0.240 (50–60%) — no clear deprivation gradient. On disability, non-disabled users (0.234) marginally outperform disabled users (0.217). None of the gaps are large enough to indicate systematic exclusion, but the gender gap in particular is worth flagging: it likely reflects the underlying course composition (STEM modules over-represented in the male sub-cohort), not a modelling choice, but I do not claim causal knowledge here. Reporting the audit is what the DEI learning objective requires; making claims about *why* the gap exists would require the intervention data OULAD does not contain.

5.10 Hyperparameter Sensitivity

evaluation/tuning_svd.csv, tuning_als.csv, and tuning_hybrid.csv record the full sweeps.

SVD. $k \in \{20, 50, 100\}$ evaluated on tune_val; NDCG@10 differences across the three ranks are small (0.159–0.167), and $k = 100$ was selected. SVD’s plateau at $k > 50$ suggests the click matrix’s effective rank is already captured by 50 factors.

ALS. 18-point grid ($n_{\text{factors}} \times \lambda \times \alpha$) on tune_val; best is ($n_{\text{factors}} = 32, \lambda = 0.01, \alpha = 1.0$) with NDCG@10 = 0.149. Increasing α *degraded* performance monotonically (from 0.149 at $\alpha = 1$ to 0.128 at $\alpha = 40$), contradicting the Hu-Koren-Volinsky music-recommendation setting where $\alpha = 40$ is the default. This is a real dataset-specific finding: OULAD click distributions have heavy positive tails per user, so treating each click as strongly confident overfits.

Hybrid. 21-point simplex ($\alpha + \beta + \gamma = 1$, step 0.2); best is (0.0, 0.8, 0.2). All configurations with $\alpha > 0.25$ underperform the CF-dropped configurations, confirming the ablation finding that the CF signal is not additive on top of content + outcome.

Time-decay. Comparing $\lambda \in \{0.0, 0.005, 0.01, 0.02\}$ (see the pre-decay run archived under `baseline_results_no_course_scoping.csv` and re-runs of `baseline_results.csv` at $\lambda = 0.01$ documented in this chapter), $\lambda = 0.01$ produces the largest gain (+2.3 P@10 points for Hybrid, +4.7 for Popularity). Both larger and smaller values regress toward the raw-click case.

5.11 Critical Evaluation and Limitations

The evaluation reaches the level of statistical confidence the marker specifically asked for: precise metric definition (Section 5.2), 95% CIs across ~17K users, paired testing with Bonferroni adjustment, five-fold temporal CV, ablation, cold-start stratification, fairness audit, and hyperparameter sensitivity across four axes. The claim I make — that GatedHybrid outperforms all six other models on the primary split and across five folds — is supported at $p < 10^{-16}$ on the four ranking-quality metrics.

The evaluation also has honest limits. First, the outcome-weighted precision is a *correlational* proxy for learning benefit, not a causal measure; without an interventional study I cannot claim that issuing these recommendations would improve outcomes. Second, ALS underperforms SVD on this dataset in a way I did not initially expect, and the CF component contributes negatively at every non-trivial weight; the “collaborative filtering” part of the “CF + content + outcome” contribution has to be reported honestly as adding little value on OULAD. Third, the cold cohort ($n = 155$) is small; the gated-hybrid cold-user finding is robust in direction but its precision estimate carries a wider CI. Fourth, the gender-fairness gap is descriptive; my project does not diagnose or mitigate it, and the final report will state that as future work.

None of these limitations undermine the primary claim. They are the honest boundaries of what an offline observational evaluation on a single dataset can support.

6 Conclusion

6.1 Summary

This project set out to build and rigorously evaluate a hybrid recommendation engine that combines collaborative filtering, feature-based content matching, and learning-outcome signals on the Open University Learning Analytics Dataset. It uses the CM3005 Data Science template: an investigation of a specific question against a real dataset, delivered as a reproducible pipeline with quantitative evaluation. Seven recommenders — Random, Popularity, truncated-SVD, Alternating Least Squares, content-based, weighted Hybrid, and a per-user Gated Hybrid — sit behind a common interface and are scored under identical conditions. The evaluation reports six metrics on a primary temporal split, five-fold temporal cross-validation, 95% bootstrap confidence intervals, paired- t tests with Bonferroni adjustment across the five metrics, an ablation study, cold-start stratification, a fairness audit across gender, IMD band, and disability, and hyperparameter sensitivity across four axes. The Gated Hybrid is the leading model on Precision@10, NDCG@10, and outcome-weighted Precision@10 on the primary split, and its lead survives cross-validation. The relative gain over the preliminary-report baseline is 18% on Precision@10.

6.2 Contribution

The contribution is not a new algorithm. It is a *systematic combination* of established techniques — collaborative filtering, feature-based content filtering, and outcome-aware evaluation — implemented and evaluated end-to-end on a publicly available educational dataset, with enough methodological rigour that the numbers survive independent scrutiny. Two specific elements are load-bearing. First, the outcome-weighted precision metric is precisely defined (Section 5.2), validated against a uniform-outcomes baseline that reduces it to standard precision, and computed strictly from training data so no test-window information leaks into the metric. Second, the seven-model progression from Random through Gated Hybrid exposes the contribution of each component — content dominates, outcome adds ~ 1 point, collaborative filtering contributes negatively at any non-trivial weight — with paired- t significance at $p < 10^{-16}$ on $\sim 17,000$ users. Together, the metric and the progression let a marker distinguish the *engineering* claim (that the pipeline works and is reproducible) from the *empirical* claim (that on OULAD, outcome-aware content filtering with a cold-start gate is the strongest offline recommendation strategy tested).

6.3 Broader Themes

Three themes deserve stating. The first is that ablation-driven design is worth doing explicitly. The Gated Hybrid did not appear in the preliminary design; it emerged from the cold-start stratification, which itself was only added because the marker’s feedback on the preliminary report asked for stratified reporting. What began as a hedge against a rubric criticism turned into the project’s best-performing recommender. Reporting the ablation honestly is what let the design decision surface, and this is a pattern I will use in future projects.

The second theme is the tension between recommendation quality and popularity bias. Recommenders that maximise Precision@K on OULAD concentrate on the head of the item distribution — Gini coefficient 0.75 for the Gated Hybrid versus 0.30 for Random. Content and outcome signals do not fix this by themselves; they *amplify* the head because the head is where the strong outcome-correlated resources live. For an educational recommender this is not automatically bad — course-central resources are course-central for a reason — but it is worth surfacing as a design trade-off rather than a rubric line-item, because the choice of how to trade quality for diversity should sit with the pedagogical stakeholder, not with the recommender.

The third theme is that “canonical” methods do not transfer without empirical checking. Alternating Least Squares with confidence weighting is *the* implicit-feedback CF method (Hu, Koren & Volinsky, 2008), and I expected it to dominate SVD on this dataset. Instead it underperformed at every confidence-alpha value tried, because OULAD click distributions have heavy positive tails per user (browsing behaviour) that treating each click as strongly confident overfits. This is a finding that would not have surfaced without running the grid; it is also a caution against assuming that any method wins in a new domain simply because it wins in the paper’s original domain.

6.4 Limitations

Four limitations are worth restating. First, the outcome-weighted precision is a *correlational* proxy for learning benefit; without an interventional study I cannot claim that issuing these recommendations would improve outcomes. Second, the cold cohort ($n = 155$) is small, so the Gated Hybrid’s cold-user precision estimate carries a wider confidence interval than the aggregate. Third, the gender-fairness gap in Section 5.9 is descriptive: my project reports it but does not diagnose or mitigate it. Fourth, the service layer (FastAPI + Streamlit) is designed

in Chapter 3 but not implemented at the draft stage; the twenty-week workplan (§3.8) schedules delivery in Weeks 16–17 (20 August – 2 September) before the 28 September 2026 final submission.

6.5 Future Work

The most direct extension is a learning-to-rank re-rank stage (LambdaMART on the union of component scores plus item and user features), which is the industry-standard hybrid pattern and would provide a legitimate ceiling comparison against the current linear ensemble. A grade-prediction diagnostic — correlating each model’s per-user precision with the eventual `final_result` in OULAD — would connect the offline OWP metric to a longitudinal outcome, tightening the “learning benefit” claim without requiring an interventional study. A knowledge-tracing component (BKT or DKT) could enter the hybrid as a fourth signal representing modelled learner state, which would let the evaluation directly compare “knowledge-tracing-flavoured” and “content-and-outcome-flavoured” recommendations on the same footing. Beyond OULAD, cross-dataset generalisation to a MOOC log (e.g. MITx / HarvardX) would test whether the “content dominates, CF contributes negatively” finding is a property of OULAD’s course-scoped catalogue or a more general educational-recommender fact. Finally, the fairness gap warrants its own investigation with the specialist literature on fairness-in-recommendation as its starting point — the numbers reported here are the input, not the conclusion, of that investigation.

References

1. Bousbahi, F. and Chorfi, H., 2015. MOOC-Rec: A case based recommender system for MOOCs. *Procedia — Social and Behavioral Sciences*, 195, pp.1813–1822.
2. Burke, R., 2002. Hybrid recommender systems: Survey and experiments. *User Modeling and User-Adapted Interaction*, 12(4), pp.331–370.
3. Corbett, A.T. and Anderson, J.R., 1995. Knowledge tracing: Modeling the acquisition of procedural knowledge. *User Modeling and User-Adapted Interaction*, 4(4), pp.253–278.
4. Drachsler, H., Verbert, K., Santos, O.C. and Manouselis, N., 2015. Panorama of recommender systems to support learning. In: F. Ricci, L. Rokach and B. Shapira (eds.), *Recommender Systems Handbook*, 2nd edn., pp.421–451. Springer, New York.
5. Hlosta, M., Zdrahal, Z. and Zendulka, J., 2017. Ouroboros: Early identification of at-risk students without models based on legacy data. *Proceedings of the Seventh International Learning Analytics & Knowledge Conference (LAK '17)*, pp.6–15.
6. Hu, Y., Koren, Y. and Volinsky, C., 2008. Collaborative filtering for implicit feedback datasets. *Proceedings of the Eighth IEEE International Conference on Data Mining (ICDM)*, pp.263–272.
7. Koren, Y., Bell, R. and Volinsky, C., 2009. Matrix factorization techniques for recommender systems. *IEEE Computer*, 42(8), pp.30–37.
8. Kuzilek, J., Hlosta, M. and Zdrahal, Z., 2017. Open University Learning Analytics dataset. *Scientific Data*, 4(1), pp.1–8.
9. Manouselis, N., Drachsler, H., Vuorikari, R., Hummel, H. and Koper, R., 2011. Recommender systems in technology-enhanced learning. In: F. Ricci, L. Rokach, B. Shapira and P.B. Kantor (eds.), *Recommender Systems Handbook*, 1st edn., pp.387–415. Springer, Boston.

10. Piech, C., Bassen, J., Huang, J., Ganguli, S., Sahami, M., Guibas, L.J. and Sohl-Dickstein, J., 2015. Deep knowledge tracing. *Advances in Neural Information Processing Systems*, 28, pp.505–513.
11. Rendle, S., Freudenthaler, C., Gantner, Z. and Schmidt-Thieme, L., 2012. BPR: Bayesian personalized ranking from implicit feedback. *Proceedings of the Twenty-Fifth Conference on Uncertainty in Artificial Intelligence (UAI)*, pp.452–461.
12. Ricci, F., Rokach, L. and Shapira, B. (eds.), 2015. *Recommender Systems Handbook*, 2nd edn. Springer, New York.
13. Rienties, B., Boroowa, A., Cross, S., Kubiak, C., Mayles, K. and Murphy, S., 2016. Analytics4Action Evaluation Framework: A review of evidence-based learning analytics interventions at the Open University UK. *Journal of Interactive Media in Education*, 2016(1), Article 2, pp.1–11.
14. Thai-Nghe, N., Drumond, L., Krohn-Grimberghe, A. and Schmidt-Thieme, L., 2010. Recommender system for predicting student performance. *Procedia Computer Science*, 1(2), pp.2811–2819.
15. Wilson, K.H., Karklin, Y., Han, B. and Ekanadham, C., 2016. Back to the basics: Bayesian extensions of IRT outperform neural networks for proficiency estimation. *Proceedings of the 9th International Conference on Educational Data Mining*, pp.539–544. arXiv:1604.02336.
16. Zawacki-Richter, O., Marín, V.I., Bond, M. and Gouverneur, F., 2019. Systematic review of research on artificial intelligence applications in higher education — where are the educators? *International Journal of Educational Technology in Higher Education*, 16(1), Article 39, pp.1–27.